

Macroinvertebrate indicators of ecological status in Mediterranean temporary stream types of the Balearic Islands

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ABSTRACT

In the last years the Water Framework Directive (WFD) has encouraged the development of classification systems to assess the ecological status in permanent rivers and streams, but in temporary streams development has been scarce. Here, we constructed an Invertebrate Multimetric index (INVMIB) to assess the ecological status of temporary streams in Mediterranean Islands based on benthic invertebrates following the WFD guidance. We studied 60 temporary streams in different seasons ($n=270$ samples), corresponding to three different stream types that were a priori established using topographical and valley information: lowland, canyon and mountain streams. We collected invertebrates with a multihabitat-based macroinvertebrate sampling standardised for a sampling area of 2.5 m^2 , and stream water to analyse most relevant water chemistry variables. The analyses of benthic macroinvertebrate assemblages confirmed our stream typology (ANOSIM, $R=0.363$, $p<0.001$). The reference community of lowland streams was defined by taxa belonging to the order Diptera, the gastropod *Ancylus fluviatilis*, and several beetles and EPT taxa. Canyons showed a high diversity strongly dominated by dipterans and aquatic mites together with mayflies. Finally, the amphipod *Echinogammarus sicilianus-monomerus*, many dipterans of the Chironomidae family, together with beetles, mayflies, stoneflies and caddisflies were highly represented in the reference communities from mountain streams. The developed INVMIBs consisted of a combination of individual metrics that included diversity, tolerance/sensitivity and composition measures. The metrics and developed type-specific multimetric indices responded significantly to the gradients of pressures obtained from PCAs (i.e., general and organic degradation), as an integrated response to the physico-chemical variables, hydromorphological parameters and land uses. Indeed, the classification system obtained with the INVMIBs reflected the ecological quality of temporary streams of the Balearic Islands. The multimetric indices are considered a valid and accurate tool for the assessment of temporary Mediterranean streams, being currently used by the water authorities of the Balearic Islands.

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1. Introduction

The implementation of the European Water Framework Directive (WFD; European Commission, 2000) plays a key role in the conservation of freshwater ecosystems, by promoting the development of classification systems to assess the ecological status of European rivers and streams. One of the keystones of the WFD classification systems is the identification of the reference condition, defined as the state where there are no, or only very minor, deviation of the biological, physicochemical and hydromorphological characteristics of the surface water body from those observed in the same stream type under undisturbed conditions

(European Commission, 2000). The reference condition constitutes the benchmark against which to evaluate deviations from the highest ecological quality expected in the absence of significant human-induced stress (Pardo et al., 2011, 2012). It subsequently constitutes the environmental objectives for maintaining or restore the ecological status of aquatic ecosystems (Hering et al., 2003; Borja, 2005; Buffagni and Furse, 2006; Erba et al., 2009; Nöges et al., 2009). Moreover, the biological characteristics of reference streams vary according to their biogeographic and geomorphological features (Verdonschot and Nijboer, 2004). Thus, type-specific biological reference conditions, that represent the values of the biological quality elements for that surface water body type at high ecological status, shall be established.

The main purpose of the development of a water body typology is to enable type-specific reference conditions to be defined. Among the obligatory environmental factors used to characterise water body types, the geographical position is relevant due to its

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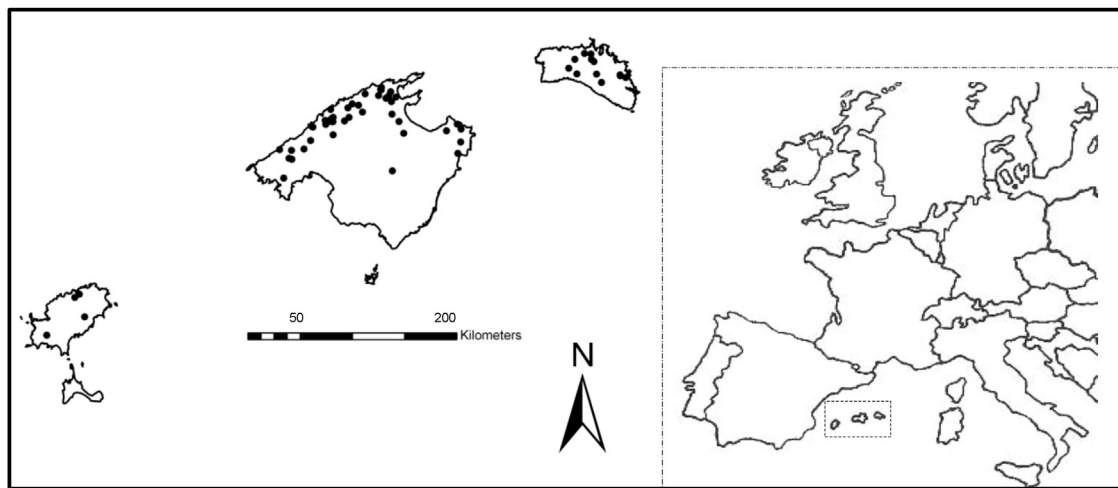


Fig. 1. Localisation and distribution of the 60 sites sampled in three of the Balearic Islands during this study.

direct relationship with the biogeography of aquatic species. In this sense, the WFD promoted the use of the ecoregions map – adapted from Illies (1967/1978) zoogeographical delineations – to facilitate a standardised surface water typology for ecological assessment and a reporting scheme for all European inland waters (European Commission, 2000). This present concept of ecoregion is too broad for an effective ecological data-based management of systems at small biogeographical scales, and smaller fluvial ecoregions should be established (Moog et al., 2004; Hughes, 2005). In addition, the case of the Island regions should be particularly assessed due to their unique ecological conditions and historical effects (Hughes, 2005).

The Mediterranean climate is characterised by sudden episodic heavy rains that generally take place during the autumn and spring months with the driest conditions prevailing during summer, which implies a periodical and predictable seasonal flood-drought sequence (Williams, 1996; Gasith and Resh, 1999; Lake, 2003; Morais et al., 2004; Pardo and Álvarez, 2006). The geology and particular precipitation patterns determine the nature of the inland waters in the Balearic Islands to be temporary streams. The effect of these hydrological events and its direct or indirect influence on other environmental parameters such as food resources, and water temperature, determines the structure and function of invertebrate communities in Mediterranean streams (Langton and Casas, 1999; García et al., 2008; Alvarez and Pardo, 2009). In particular, the temporary streams show a high natural spatial and temporal variability on their invertebrate assemblages (Alvarez and Pardo, 2007; Munné and Prat, 2011; Sánchez-Montoya et al., 2011), which makes necessary to establish a clear distinction between the influence of human impacts and typological differences in order to properly assess their ecological status.

A diversity of biologic indicators including macrophytes, fish and diatoms have been used to assess the water quality in European rivers and streams (Schaumburg et al., 2004; Breine et al., 2007; Delgado et al., 2010, 2012), but the most widely used are those based on the invertebrate community (Metcalf, 1989; Hering et al., 2003; Brabec et al., 2004; Ofenböck et al., 2004; Bennett et al., 2011; Birk et al., 2012). Indeed, benthic macroinvertebrates have proved to be suitable bioindicators in responding to multiple stressors, based on their ability to respond to a variety of environmental variables (Morais et al., 2004; Hering et al., 2006a; Johnson et al., 2007; Pardo et al., 2014).

The present approach promotes the importance of the ecological response of invertebrate assemblages to multiple pressures, allowing the assessment of the ecological status of streams in islands,

as required by the WFD (European Commission, 2000). Therefore, we combined individual metrics in a multimetric index based on its individual capacity to respond to multiple environmental pressures (Hering et al., 2006b). The main objectives of this study were (i) to identify and compare macroinvertebrate communities across the different temporary stream types existing in the Balearic Islands: lowland, canyon and mountain streams, (ii) to assess the response of the biotic assemblages along quantified gradients of human pressures, (iii) to develop a multimetric index and a classification system that accurately respond to the existing pressures, that fulfils all obligatory biological parameters of WFD normative definitions, and (iv) to assess the relevance of macroinvertebrates as indicators of the ecological status of temporary streams.

2. Materials and methods

2.1. Study area

This study was conducted in the archipelago of the Balearic Islands (Spain) that is composed by five islands located in the Western part of the Mediterranean Sea (Fig. 1). The Balearic Islands are included in the Iberic-Macaronesian ecoregion (Annex XI, European Commission, 2000). They are influenced by the Mediterranean climate and the discharge regime of these streams exhibits a strong seasonal and annual variability. These islands have a karst geology and large amounts of water stored in subterranean aquifers. Thus, the hydrological system of the Balearic Islands is marked by its climate and geomorphology, configuring the existence of temporary streams, denoted “torrents”, which during the hot and dry summer cease to flow, except for those sustained by springs that are present in the north of Majorca Island (Delgado et al., 2013).

The Balearic Islands comprise a wide spatial variation due to its topography and local geomorphology. Lowland streams are the most abundant type, located at low altitude and close to the most populated areas. The other stream types (canyon and mountain) are located only in the North of Majorca Island within the Tramuntana mountain range. Canyons are deep valleys between cliffs carved in the mountains, isolated and with high slopes, while mountain streams are located in the mountains having intermediate slopes.

2.2. Field sampling and laboratory processing

The sampling design included minimally disturbed sites (*sensu* Stoddard et al., 2006) in a spatial network used as reference sites (Pardo et al., 2011, 2012) and other stream sites influenced by

Table 1

Temporary streams sampled in this study, indicating the number of reference and non-reference sites per stream type and major environmental features to characterise these stream types (i.e., range of elevation, slope, total catchment and water permanence). The number of samples that correspond to reference and non-reference sites for each stream type is also shown between brackets.

	Stream type		
	Lowlands	Canyons	Mountains
Reference sites	2 (10)	4 (23)	6 (32)
Non-reference sites	30 (123)	5 (20)	13 (62)
Total	32	9	19
Altitude (m)	6–93	39–333	23–314
Slope (%)	0.2–2.3	2.2–13.9	1.3–10.9
Catchment (km ²)	5.4–151.1	7–27.8	6.9–35.1
Water permanence (months)	5–12	5–12	8–12
Island	Majorca	Majorca	Majorca, Minorca & Ibiza

different pressures. A total of sixty sites were sampled on the three largest Balearic Islands: Majorca, Minorca and Ibiza, including reference and non-reference sites for the different stream types (Table 1). Sampling was conducted in 2005, 2006 and 2008 and the study design specially avoided sampling during initial phases of water resumption and final phases of water reduction, ensuring the existence of flowing water for a minimum period of 1 month before sampling. The 6 sampling campaigns took place in November–December (autumn 2005), February–March (winter 2006, 2008) and April–May (spring 2005, 2006 and 2008). All the studied temporary streams had a flow period greater than 6 months (Table 1) ensuring the existence of the characteristic late successional communities (Delgado et al., 2012). Each site was sampled in at least two different seasons, resulting in a total of 270 samples.

Physico-chemical variables were measured in situ with portable metres: water temperature and pH (Thermo Orion 290+), electric conductivity at 25 °C (Thermo Orion 115A+) and dissolved oxygen (WTW Oxi 197 oxymeter). Water samples were collected from the streams into polypropylene bottles, stored in dark conditions at 4 °C and transported to the laboratory. Water chemical analysis was carried out by following standard methods (APHA, 1989): BOD₅ was measured with an oxitop WTW after 5 days at 20 °C; alkalinity by the potentiometric method; nitrates (N.NO₃⁻), ammonium (N.NH₄⁺), silicon dioxide (SiO₂), phosphates (P.PO₄³⁻) using an auto-analyzer for nutrients quantification (Auto-Analyzer 3, Bran + Luebbe, Germany); calcium (Ca²⁺), magnesium (Mg²⁺), potassium (K⁺), sodium (Na⁺) with a spectrophotometer of masses; while chlorides (Cl⁻) and sulphates (SO₄²⁻) were measured with the colorimetric procedure of ICP (inductively coupled plasma).

Table 2

Summary of environmental variables within reference and non-reference sites studied in each stream type (mean ± SD). The number of samples that correspond to reference and non-reference sites for each stream type is also shown between brackets.

	Reference sites			No-reference sites		
	Lowlands (N = 10)	Canyons (N = 23)	Mountains (N = 32)	Lowlands (N = 123)	Canyons (N = 20)	Mountains (N = 62)
Conductivity (μS/cm)	747.86 ± 29.46	387.44 ± 41.33	753.68 ± 55.5	1573.93 ± 64.12	515.48 ± 36.68	838.07 ± 43.18
pH	7.66 ± 0.12	8.17 ± 0.09	8.04 ± 0.06	7.86 ± 0.03	7.96 ± 0.11	7.79 ± 0.05
Oxygen (mg/l)	9.36 ± 0.56	10.15 ± 0.51	9.65 ± 0.19	8.34 ± 0.28	9.32 ± 0.33	8.22 ± 0.38
N.NO ₃ (mg/l)	0.32 ± 0.13	0.34 ± 0.13	0.54 ± 0.12	2.96 ± 0.42	0.52 ± 0.1	1.03 ± 0.15
N.NH ₄ (mg/l)	0.03 ± 0.02	0.02 ± 0.01	0.02 ± 0.01	0.39 ± 0.12	0.02 ± 0.01	0.19 ± 0.1
P.PO ₄ (mg/l)	0.01 ± 0.01	0.02 ± 0.01	0.02 ± 0.0	0.43 ± 0.09	0.02 ± 0.0	0.27 ± 0.09
Chlorophyll <i>a</i> (mg/m ²)	5.92 ± 2.23	9.01 ± 2.44	7.24 ± 1.25	20.55 ± 1.92	9.58 ± 1.48	19.05 ± 2.48
AFDM (g/m ²)	6.18 ± 1.07	9.88 ± 2.14	7.73 ± 1.83	15.4 ± 1.36	15.28 ± 3.01	13.49 ± 1.42
Artificial areas (%)	0.0 ± 0.0	0.66 ± 0.25	0.0 ± 0.0	1.61 ± 0.17	1.32 ± 0.34	2.34 ± 0.27
Agricultural areas (%)	18.01 ± 3.78	6.92 ± 2.49	6.33 ± 0.45	53.2 ± 1.72	24.95 ± 1.14	27.38 ± 0.8
Forest and seminatural areas (%)	81.99 ± 3.78	92.42 ± 2.74	93.67 ± 0.45	45.14 ± 1.76	73.73 ± 1.41	70.18 ± 0.87
HQA score	36.0 ± 2.67	38.94 ± 0.41	45.8 ± 1.8	34.78 ± 0.92	32.69 ± 2.31	38.19 ± 1.3
HMS score	2.0 ± 0.67	12.0 ± 3.9	15.33 ± 3.7	34.61 ± 2.7	21.65 ± 4.17	31.6 ± 2.98

Discharge (L/s) was calculated by multiplying the stream cross sectional area by the integrated mean velocity ($n = 3$) recorded at fix stream positions per site and sampling time. Periphyton samples were taken from the upper surfaces of three stones (20–60 cm²) by scraping with a toothbrush and rinsing after with distilled water, being processed in the laboratory to quantify chlorophyll *a* (Chl *a*) and the periphytic biomass as the ash-free dry mass (AFDM). For further information regarding the laboratory procedure of periphyton samples see Delgado et al. (2012). Hydromorphological parameters were registered by following the CARAVAGGIO (Core Assessment of River hAbitat VALue and hydro-morpholoGical cOndition) protocol, along 500 m upstream on each site during 2006. This protocol represents a version of the River Habitat Survey (RHS) specifically modified and implemented to cover the characteristics used to evaluate the hydromorphological quality of South European rivers. It allows the calculation of different indices such as HQA (Habitat Quality Assessment) and HMS (Habitat Modification Score) that indicate the availability of habitats and the degree of channel alteration, respectively (Buffagni et al., 2004; Buffagni and Kemp, 2002).

Benthic macroinvertebrate samples were collected by following a multi-habitat procedure (adapted from EPA, Barbour et al., 1999) with a kick-net (500 μm-mesh size). A total of 20 “sample units” distributed in a proportional number to the habitats present in 100 m length were sampled (total sampled area of 2.5 m²). Field samples were preserved with ethanol (70%) in plastic bags, transported to the laboratory and stored until their treatment. Invertebrate samples were washed and sorted under tap water in three different fractions (5 mm, 0.5 mm and 0.1 mm), and specimens were counted and identified under 40× magnification (Olympus U-TV1X) to the lowest possible identification level. Sub-sampling was made to obtain a representative fraction of the total community in the finer fractions when needed (Wrona et al., 1982).

2.3. Reference conditions

Information on land cover in the Balearic Islands was extracted from CORINE (COordination of INFORMATION of the Environment) Land Cover 1990 (Bossard et al., 2000), to calculate the percentage of land use in the catchment area per sampling site. Databases from the regional government of the Balearic Islands were used to check for point sources of pollution (e.g., existence of water treatment plants) and other human disturbances in the Balearic streams. The reference sites had to satisfy a series of a priori selection criteria based on the absence of significant pressures (Wallin et al., 2003; Pardo et al., 2011, 2012), considering the percentages of artificial and agricultural land uses, absence of sewage effluents, no

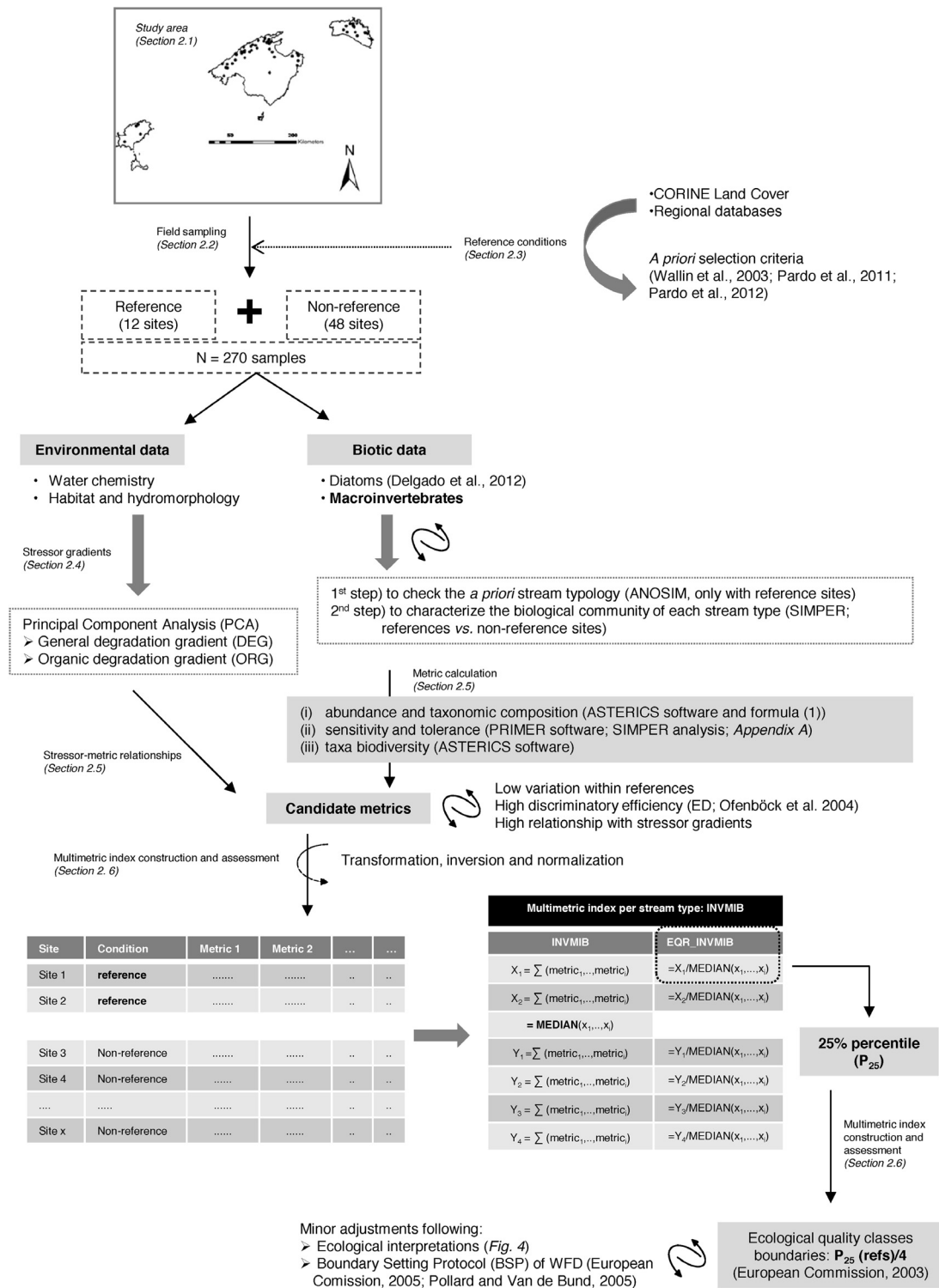


Fig. 2. Flow diagram showing the main steps followed to develop the multimetric index of Mediterranean temporary streams. Literature references and sections from this paper describing each step of the procedure were given into brackets.

hydromorphological alterations of the stream bank and absence of significant flow regulation. In this study, we used restricted pressure levels for artificial (0% of catchment area, although one site had 2% due to the presence of a town in its small catchment, but without inputs to the stream) and agricultural surfaces ($\leq 7\%$ of catchment area, one site in lowland streams had 29%, and other site in canyon streams had 22%, both percentages corresponding to non irrigated land with high levels of patchy natural vegetation,

Table 2), following the conclusions of the review on European reference thresholds by Pardo et al. (2011). A posterior check with water quality and macroinvertebrate communities was performed to ensure the absence of impact from non-evaluated pressures at the reference sites (Pardo et al., 2012). The main steps followed during this study were summarised in a flow diagram (Fig. 2) from data collection to the final multimetric index and ecological classification system.

The invertebrate community was analysed with PRIMER software v. 6 (Plymouth Routines in Multivariate Ecological Research) and CANOCO 5.0 for Windows. Abundance data were previously transformed by $\log(x + 1)$ to prevent distortions caused by the most abundant taxa. In a first step, the invertebrate community of reference sites was used to check the geomorphological stream typology predefined for the Balearic Islands (lowland, canyon and mountain streams) by an ANalysis Of SIMilarities (ANOSIM, with 999 permutations, a significance level of 0.1% and pairwise tests), which is a non-parametric test to compare different groups based on a similarity distance measure. In a second step, the degree of similarity and dissimilarity within and between different groups (i.e., reference vs. non-reference sites) was calculated by a SIMPER (SIMilarity PERcentage) analysis. This method allowed us to obtain the biological community that characterised each stream type and to perform the calculation of metrics. These analyses were based on Bray–Curtis similarity that combines information on the species abundances and the occurrence of species in particular groups.

2.4. Stressor gradients

The non-reference sites were influenced by different degrees of human pressures, aimed to cover the whole gradient of degradation to which the temporary streams are subjected in the Balearic Islands. Environmental variables collected during this study (i.e., water permanence, alkalinity, nutrients, discharge, dissolved oxygen, electric conductivity, chlorophyll *a*, AFDM, hydromorphological indices and land uses (e.g., artificial, agricultural and forestry uses)) represented a wide range of stressors that may have a potential strong impact on the invertebrate communities. These variables were explored with PCAs (Principal Component Analyses) to reduce the information of all environmental variables into different stressor gradients: (i) a multiple degradation gradient (DEG), with a matrix of 270 samples $\times$ 18 parameters, that will integrate the whole information and variation from different pressures in all stream types, and (ii) an organic degradation gradient by stream type (i.e., organic degradation = ORG: oxygen demand, nutrient concentrations, dissolved oxygen, discharge and forestry land use in the basins), with a matrix of 133 samples $\times$ 6 parameters for lowlands, 43 samples $\times$ 6 parameters for canyons and 94 samples $\times$ 6 parameters for mountain streams. Data were previously standardised (by the mean and the standard deviation) prior to run the analysis.

2.5. Metric calculation and stressor-metric relationships

Individual metrics were selected following the requirements of the WFD normative definitions: (i) the abundance and taxonomic composition of the invertebrate community, (ii) the sensitivity or tolerance of taxa to stressors, and (iii) taxa biodiversity. Invertebrate taxa abundance was used to calculate individual metrics, such as Bray Curtis index (BC). BC reflects differences between two samples with differing community composition and/or total abundance. We estimated the BC index between reference and non-reference sites by calculating the median of the abundance of the invertebrate at reference sites and then calculated the BC index from the latter to non-reference sites. The sensitive/tolerance genera metric was calculated using the similarity percentage analysis (SIMPER) that examines the contribution of each taxon to the average resemblance among reference and non-reference sites (Delgado et al., 2012; Lucena-Moya and Pardo, 2012). Therefore, taxa exclusively present at the reference sites and that contributed to 90% of the similarity within reference group were considered to be sensitive taxa. Meanwhile, taxa characteristics for the non-reference sites were identified as tolerant or disturbed taxa. Finally, the relative abundance of sensitive and tolerant taxa, and others

indices such as taxon richness, diversity and EPT taxa were calculated with the ASTERICS program (software v.3.01). These analyses were performed at the general taxonomic genus level (except for Oligochaeta and Diptera that were identified to family and tribu) for each stream type separately.

After these calculations, all metrics obtained were correlated with single environmental variables and also with the extracted scores from PCAs. Spearman correlations were performed with the SPSS software v.14.0 to assess the relationship between metrics and the stressor gradients and to identify significant responses with potential to be indicators of the ecological status.

2.6. Multimetric index construction and assessment

Whisker-plots and ANOVA or *U* Mann–Whitney tests, depending on the parametric character of the data, were applied to each individual metric (and also to each multimetric index, see below) to test for significant differences and to discriminate between reference and non-reference sites. The discriminatory efficiency (DE) was calculated as the percentage of stressed samples with metric values lower than the 25th percentile of reference values for decreasing metrics, and higher than the 75th percentile for increasing metrics, respectively (Ofenböck et al., 2004).

Those metrics showing low variability within reference sites, high relationships with stressor gradients and the highest DEs between reference and non-reference sites were considered as the candidate metrics. Abundance metrics were previously transformed by $\log(x + 1)$ to prevent distortions caused by the most abundant taxa, while percentage metrics were expressed as a fraction of unity. Before the combination of metrics, the response of each candidate metric and its magnitude was explored. Therefore, metrics had to respond to the gradient of pressure in the same direction; if they did not, they were inverted (e.g., PABST). Moreover, they also should have the same scale, so all candidate metrics were standardised by dividing them by the same number (i.e., the median of the reference sites) before summed in a multimetric index per stream type, denoted as INVMIB (Invertebrate Multimetric index of Balearic Islands). The median value from the distribution of reference sites is considered the most robust value to be used as the reference value in the classification of the ecological status (relatively few data/sites needed for sufficient confidence; European Commission, 2003).

The ecological status, defined by the biological community, was expressed as EQR (Ecological Quality Ratio, here denoted as EQR_INVMIB), which was obtained by dividing the multimetric index for a site by the median value of the multimetric index obtained from the reference population of sites per stream type, resulting in EQR values comprised between 0 and >1. Then, we used the 25th percentile (P_{25}) of the EQR reference values to identify the “High”–“Good” boundary of the classification, and the remaining quality class boundaries were defined in equal bands. We focused on metrics responding linearly to disturbance to improve applicability. We used this approach based on the best linear relationships achieved between the EQR_INVMIB values and the range of environmental alteration (i.e., PCA axis I DEG), and by following the recommendations from the Boundary Setting Protocol (BSP) of WFD (European Commission, 2005; Pollard and van de Bund, 2005). To determine the width of the four remaining classes (i.e. Good, Moderate, Poor and Bad), the P_{25} was divided by four ($P_{25}/4$) and the resulting number was subtracted consecutively from the previous boundary limit to get the next ones (European Commission, 2003). The result of this was that equidistant ecological classes were obtained. This approach was modified following the ‘Guidance on the inter-calibration process 2008–2011’ (Schmedtje et al., 2009), a document which aims to confer a better ecological meaning to the previous band widths obtained simply by

mathematical calculations. According to that, the ecological-class boundary limits or class centres can be located using either a metric that shows discontinuity along a pressure gradient, or at least two metrics that respond in different ways to the pressure (i.e. metrics cross, as one metric responds positively to the pressure while the other responds negatively). The crossing between metrics can be related to a class boundary or a class centre. This procedure was followed for each stream type, and the results were averaged to obtain one single value for each ecological-class boundary limit.

3. Results

3.1. Reference conditions

The biggest island of Balearic Islands is Majorca and had the highest number of sampling sites, followed by Minorca and Ibiza. A total of 12 sites were designated as reference: two in lowlands, four in canyons and six in mountain stream types, all of them in Majorca Island (Table 1). The global ANOSIM test showed that there were significant differences ($R = 0.363$, $p < 0.001$, 999 permutations) in the reference assemblage composition among the stream types defined a priori. All the stream types were significantly distinct one from the others (pair-wise comparisons, $p < 0.05$). The reference invertebrate community showed a high similarity per each stream type: 49.9%, 39.7% and 39.6%, for lowland, canyon and mountain streams, respectively. Moreover, the dissimilarities between stream types were always higher than 70%. For a detailed list of the taxa that most contributed to both similarity within stream types and dissimilarity among stream types (see Appendix A).

The reference community of lowland streams was defined by taxa belonging to the order Diptera (22.42%), the gastropod *Ancylus fluviatilis* or different beetles like *Halipplus*, *Laccobius* and *Agabus*. There was also a strong contribution of EPT genera (~13%) such as *Hydroptila*, *Tyrrhenoleuctra* and *Cloeon*. The reference community of canyons was strongly dominated by dipterans and aquatic mites (29.6%) together with mayflies, such as *Cloeon* and *Caenis*, which contributed to the reference similarity with a 7.2% and a 4.4%, respectively. Beetles of the Dytiscidae family, together with different taxa of dipterans (e.g., *Chironomini* and *Dasyhelea*), gastropods (e.g., *Gyraulus*), bedbugs (e.g., *Notonecta*) and caddisflies (e.g., *Hydroptila* and *Mesophylax*), contributed less, although determined the high diversity of this stream type. Differences in the water availability of these streams (i.e., some streams completely dry during summer vs. others that dry up to pools or “gorgs”) led to its wide variability, overlapping with sites grouped in the mountain stream type (see Fig. 2). Finally, the reference community of mountain streams was highly represented by the amphipod *Echinogammarus sicilianus-monomerus*. The contribution of dipterans of the Chironomidae family (17%) was closer to that registered in other stream types. Mayflies, stoneflies and caddisflies such as *Alainites*, *Baetis*, *Caenis*, *Cloeon*, *Leuctra*, *Hydroptila*, *Tinodes*, *Polycetropus* and *Rhyacophila* contributed to the high similarity within this stream type. Typical beetles such as *Hydraena* and *Oulimnius* were also abundant in the reference community of mountain streams.

Regarding the physico-chemical status, reference sites met the WFD requirements on the absence of significant pressures (Table 2). Canyons were the least impacted stream type, with the lowest values of electric conductivity and agricultural areas and the highest values of dissolved oxygen (Table 2). Electric conductivity was high in all temporary streams, due to the high natural contents of different ions in the water from the dominant calcareous geology (Table 2). Nutrient concentrations, chlorophyll *a* and biomass (AFDM) showed the highest values in the non-reference sites. Non-reference lowland streams were the most impacted ones and showed the highest nutrient concentrations as well as the highest

percentages of agricultural areas and hydromorphological alterations (Table 2).

3.2. Stressor gradients

An initial PCA analysis was performed including all the samples ($n = 270$) and 18 environmental variables that integrated the whole variation existing due to all different pressures (i.e., general degradation = DEG, Table 3). The extracted axes identified different gradients as a combination of several pressures and stressors. Axis I DEG explained 26.3%, where nitrate and agricultural land use, among other variables, showed an opposite tendency to dissolved oxygen and HQA score, indicating a gradient from well preserved and minimally impaired streams (mountains and canyons) to nutrient impaired streams such as lowland streams (Table 3). Axis II DEG explained 12.3% of the total variance, indicating a seasonal dynamic marked by the relationships between discharge and oxygen contents. High discharge levels occurring during autumn–winter were positively related with oxygen saturation.

As the stream typology was confirmed by the existence of distinct type-specific biological communities and to simplify previous interpretations, new PCAs (one per stream type) were run only with 6 environmental variables influenced by organic/nutrient inputs (ORG), which are the dominant degradation pressure in the Balearic Islands. In these PCAs, axis I and II explained together 56.8%, 55.9% and 59.8% of total variance across sites for lowlands, canyons and mountains, respectively. In general, axis I ORG was confirmed as an organic/nutrient gradient with BOD₅ and phosphates (also nitrate for canyons and mountain streams) usually opposed to dissolved oxygen and discharge, and thus separating impacted sites from less impacted and reference sites (Table 3). Axis II ORG was a temporal gradient similar to the gradient obtained for the axis II DEG. Within the axis II ORG, nitrates increased with discharge in impacted sites, although it was negatively related to forestry land uses (Table 3). Indeed, sites with highly irrigated surfaces and agricultural catchments showed a strong nutrient enrichment during periods of high discharge.

3.3. Candidate metrics and multimetric index construction

A total of 410 invertebrate taxa were identified in this study, although only 246 taxa were used in the metrics development (i.e., those taxa with relative abundances in the total database higher than 1%). The invertebrate community inhabiting the temporary streams of the Balearic Islands showed a high diversity, and it was characterised by a total of 47 sensitive and 36 tolerant taxa (Appendix B). From them, mountain streams showed the highest presence of sensitive (20) and tolerant taxa (16), while canyons showed the lowest number (12 sensitive and 5 tolerant taxa).

The candidate metrics selected in the INVMIB for all stream types complied with the normative definitions of the WFD (Table 4). The INVMIB was calculated by the combination of these candidate metrics per stream type: INV1MIB (lowlands), INV2MIB (canyons) and INV5MIB (mountains) (Table 4). Some of these metrics such as the richness of sensitive taxa (RSS) was included in all the multimetric indices, while other metrics gave reliable results only in one stream type. For example, the EPT abundance of classes was a good discriminator of impairment for mountain streams, and the abundance of Ephemeroptera resulted to be a good discriminator for canyons, due to the high representation of taxa of the family Baetidae inhabiting these streams (Table 4).

The INV1MIB responded positively to the axis I DEG (i.e., –agricultural areas and –NO₃) and to the axis I ORG (i.e., –NO₃ and +O₂). The individual metrics were similarly related to these gradients (Table 3). The BC index and the RSS metric responded

Table 3

Variable loading in axes I and II obtained from PCAs of general (DEG) and organic (ORG) degradation gradients for each stream type, and Spearman correlations among candidate metrics and the multimetric index INVMIB for each stream type (i.e., INV1MIB, INV2MIB and INV5MIB) with these axes. Total $n = 133$ (lowlands), 43 (canyons) and 94 (mountains).

Stream type	Variables	Axis I DEG	Axis II DEG	Axis I ORG	Axis II ORG
Lowlands	Water permanence (months)	−0.027	−0.071	–	–
	Alcalinity (mEq/l)	−0.241	−0.286	–	–
	P-PO ₄ (mg/l)	−0.273	−0.319	−0.578	−0.086
	N-NO ₃ (mg/l)	−0.282	0.355	0.087	−0.680
	N-NH ₄ (mg/l)	−0.241	−0.292	–	–
	DIN (mg/l)	−0.350	0.218	–	–
	SiO ₂ (mg/l)	−0.287	−0.032	–	–
	Cl (mg/l)	−0.267	−0.083	–	–
	Discharge (l/s)	−0.091	0.414	0.179	−0.546
	Oxygen (%)	0.214	0.412	0.504	−0.014
	Electric conductivity (mS/cm)	−0.338	−0.076	–	–
	Chlorophyll a (mg/m ²)	−0.204	0.283	–	–
	AFDM (g/m ²)	−0.182	0.121	–	–
	HQA score	0.151	−0.111	–	–
	HMS score	−0.173	0.093	–	–
	LRD score	0.158	−0.287	–	–
	Artificial areas (%)	−0.146	0.019	–	–
	Agricultural areas (%)	−0.334	−0.005	–	–
	Forest and seminatural areas (%)	–	–	0.243	0.481
	BOD ₅ (mg/l)	–	–	−0.560	0.005
	INV1MIB	0.496*	0.016	0.334*	0.353*
	Bray Curtis index	0.328*	−0.098	0.156	0.254*
	Percentage of abundance of tolerant taxa (PABST)	−0.287*	−0.169	−0.411*	−0.138
	Richness of sensitive taxa (RSS)	0.528*	−0.167	0.153	0.410
Canyons	Water permanence (months)	−0.027	−0.071	–	–
	Alcalinity (mEq/l)	−0.241	−0.286	–	–
	P-PO ₄ (mg/l)	−0.273	−0.319	−0.593	0.270
	N-NO ₃ (mg/l)	−0.282	0.355	−0.522	−0.299
	N-NH ₄ (mg/l)	−0.241	−0.292	–	–
	DIN (mg/l)	−0.350	0.218	–	–
	SiO ₂ (mg/l)	−0.287	−0.032	–	–
	Cl (mg/l)	−0.267	−0.083	–	–
	Discharge (l/s)	−0.091	0.414	−0.186	−0.434
	Oxygen (%)	0.214	0.412	0.098	0.400
	Electric conductivity (mS/cm)	−0.338	−0.076	–	–
	Chlorophyll a (mg/m ²)	−0.204	0.283	–	–
	AFDM (g/m ²)	−0.182	0.121	–	–
	HQA score	0.151	−0.111	–	–
	HMS score	−0.173	0.093	–	–
	LRD score	0.158	−0.287	–	–
Canyons	Artificial areas (%)	−0.146	0.019	–	–
	Agricultural areas (%)	−0.334	−0.005	–	–
	Forest and seminatural areas (%)	–	–	0.103	0.623
	BOD ₅ (mg/l)	–	–	−0.567	0.318
	INV2MIB	0.505*	−0.223	0.202	0.589*
	Abundance of Ephemeroptera (EPHEMAb)	0.361*	−0.132	0.330*	0.376*
	Margalef index	0.303*	−0.268	0.072	0.485*
Mountains	Richness of sensitive taxa (RSS)	0.531*	−0.081	0.150	0.552*
	Water permanence (months)	−0.027	−0.071	–	–
	Alcalinity (mEq/l)	−0.241	−0.286	–	–
	P-PO ₄ (mg/l)	−0.273	−0.319	0.554	−0.093
	N-NO ₃ (mg/l)	−0.282	0.355	0.249	0.251
	N-NH ₄ (mg/l)	−0.241	−0.292	–	–
	DIN (mg/l)	−0.350	0.218	–	–
	SiO ₂ (mg/l)	−0.287	−0.032	–	–
	Cl (mg/l)	−0.267	−0.083	–	–
	Discharge (l/s)	−0.091	0.414	−0.116	0.692
	Oxygen (%)	0.214	0.412	−0.483	0.163
	Electric conductivity (mS/cm)	−0.338	−0.076	–	–
	Chlorophyll a (mg/m ²)	−0.204	0.283	–	–
	AFDM (g/m ²)	−0.182	0.121	–	–
	HQA score	0.151	−0.111	–	–
	HMS score	−0.173	0.093	–	–
	LRD score	0.158	−0.287	–	–
	Artificial areas (%)	−0.146	0.019	–	–
	Agricultural areas (%)	−0.334	−0.005	–	–
	Forest and seminatural areas (%)	–	–	−0.244	−0.651
	BOD ₅ (mg/l)	–	–	0.570	−0.018
	INV5MIB	0.676*	0.113	−0.753*	−0.500*
	Bray Curtis index	0.571*	0.135	−0.637*	−0.298*
	Percentage of abundance of tolerant taxa (PABST)	−0.544*	−0.061	0.550*	0.342*
	Richness of sensitive taxa (RSS)	0.546*	0.143	−0.665*	−0.453*
	EPT index (% abundance classes)	0.616*	0.108	−0.716*	−0.494*

* Significant correlations at the level 0.05.

Table 4

Resume of the individual metrics used in the elaboration of the Invertebrate Multimetric index (INVMIB) in each stream type by indicating the response of each metric to the pressure: decrease (▼) or increase (▲), their transformations and normalizations needed to its calculation.

Stream type	INVMIB	Meaning	Response to the pressure	Transformation	Inversion	Normalisation
Lowlands (INV1MIB)	Bray Curtis	Bray–Curtis index	▼	A fraction of unity	No	Median
	PABST	Percentage of abundance of tolerant taxa	▲	A fraction of unity	Yes	
	RSS	Richness of sensitive taxa	▼	No	No	
Canyons (INV2MIB)	EPHEMab	Abundance of Ephemeroptera	▼	Log($x + 1$)	No	Median
	Margalef index	Diversity of Margalef	▼	No	No	
	RSS	Richness of sensitive taxa	▼	No	No	
Mountains (INV5MIB)	Bray Curtis	Bray–Curtis index	▼	A fraction of unity	No	Median
	PABST	Percentage of tolerant taxa	▲	A fraction of unity	Yes	
	RSS	Richness of sensitive taxa	▼	No	No	
	EPT index	EPT % abundance of classes	▼	A fraction of unity	No	

positively while the PABST metric showed the opposite response, as expected. The DEs of lowland streams varied between 82% for the PABST metric and 93% for the RSS metric, while for the multimetric index was 92%, with no overlap between reference and non-reference sites (Fig. 3). The INV1MIB and all of the individual metrics of the reference sites were significantly different from those of the non-reference sites (Fig. 3). In canyon streams, the INV2MIB and also the individual metrics responded positively to the axis I DEG (i.e., $-\text{NO}_3$ and $+\text{O}_2$) and to the axis II ORG (i.e., +forestal areas

and $-\text{NO}_3$). The DEs of canyon streams varied between 26% and 37% in the individual metrics, while the DE of the multimetric index was 35% (Fig. 3). The INV2MIB and all its individual metrics were significantly different between reference and non-reference sites, except for the Margalef diversity that was strongly reduced in the worst status classes (see below). The INV5MIB (mountain streams) and its individual metrics showed the strongest responses to both stressor gradients (Table 3). Moreover, the DEs varied between 53% (BC) and 68% (RSS), being 60% for the multimetric index (Fig. 3).

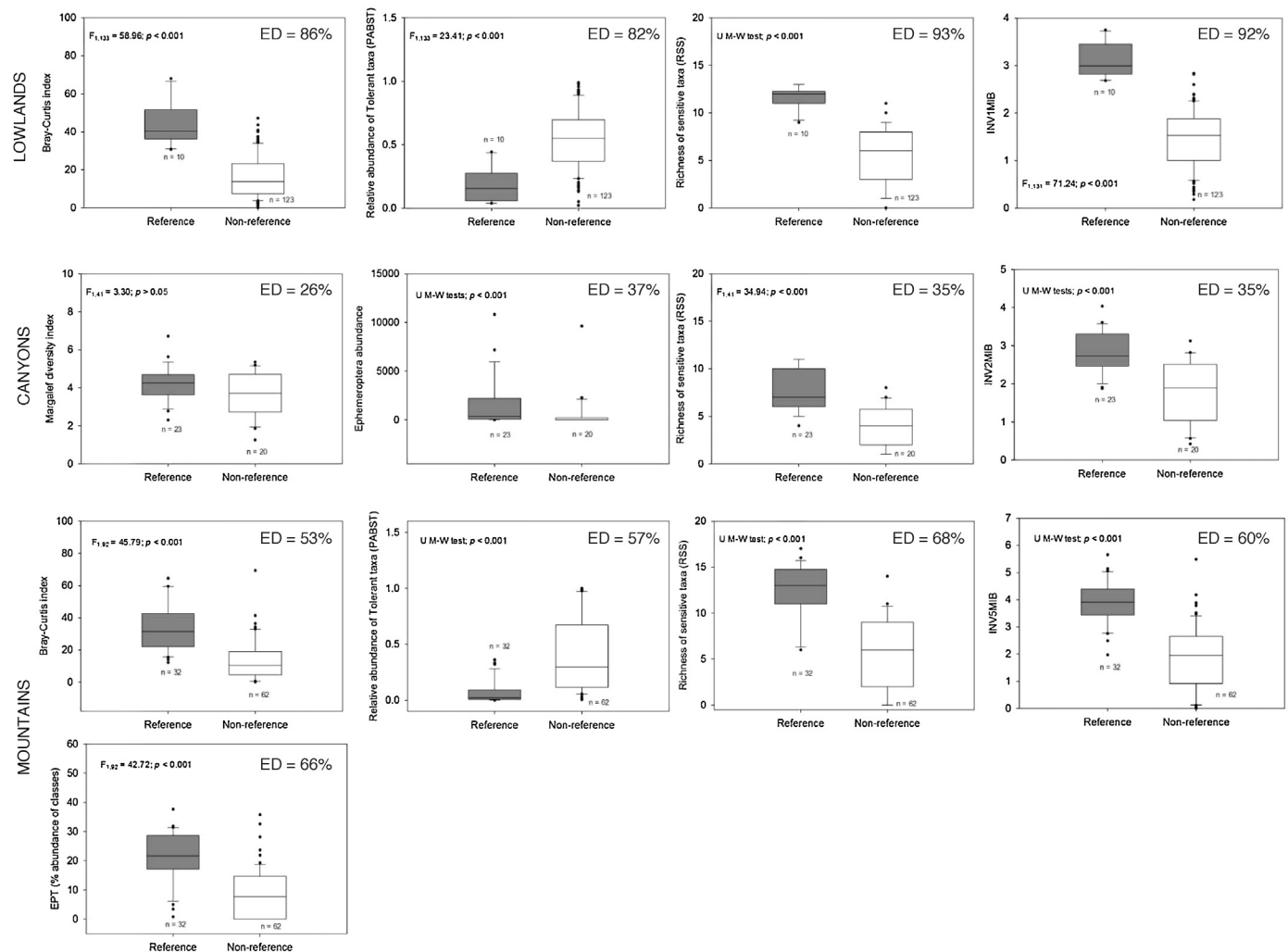


Fig. 3. Box-plot diagrams showing the discriminatory power between the reference and non-reference sites of the individual metrics for each stream type. The median values (horizontal central line), 25th and 75th percentile values (box), the maximum (upper whisker), minimum (lower whisker), extreme. Values and atypical values (*) as well as the number of samples, discriminatory efficiency (DE) and significance levels are shown in these diagrams.

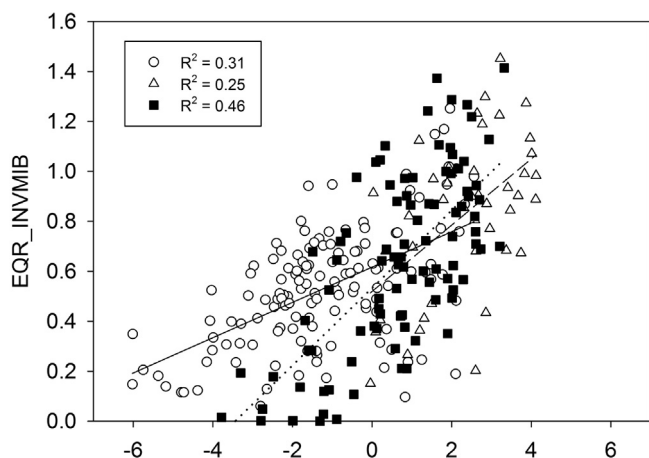


Fig. 4. Linear regressions between the EQR_INVMB of each stream type and the general degradation gradient (axis I DEG). R -squares are given for each stream type ($p < 0.001$). Lowland streams (white circle), canyons (white triangle) and mountain streams (dark square).

Similar to the other stream types, the INV5MB and its multimetric index for reference sites were significantly different for those of the non-reference sites (Fig. 3).

3.4. Multimetric index assessment

The response of the EQR to the general degradation gradient (axis I DEG) showed to be linear for all stream types (Fig. 4). The resulting boundary limits varied slightly among stream types (0.955 for lowlands, 0.914 for canyons and 0.912 for mountain streams).

The relationship between each EQR_INVMB, using the EQR values as a surrogate of the gradient of pressure, and its component metrics indicated that those metrics expected to increase with human pressures increased and metrics expected to decrease with human pressures decreased (Fig. 5). This approach allowed to quantify along the linear continuity of metrics at which values of the gradient the metrics cross in each stream type (Fig. 5). Indeed, in lowland streams, when the sensitive taxa decreased below 50% and the tolerant taxa increased more than 50%, the conditions fitted to the lower boundary of the “Moderate” class (Fig. 5A). Moreover, the crossing point between the Bray–Curtis similarity and the tolerant taxa can be identified as a point from which early deterioration occur in the gradient (Fig. 5A and C), as the percentage of tolerant taxa increased up to 30% and there was a shift from “Good” towards “Moderate” class. In all stream types, those sites located in the “Poor” and “Bad” classes showed biotic communities somehow different from that of the reference condition, with high percentages of tolerant taxa and low values of those metrics indicative of richness, diversity and sensitive taxa (Fig. 5). In mountain streams, the cross point between sensitive and tolerant taxa occurred below the boundary of the “Poor” class (Fig. 5C).

Later, and once realised that class boundaries derived with the same procedure for the different stream types were very similar, and after checking the interpretation of Fig. 5A–C, a final set of common boundaries was established in 0.93, 0.73, 0.50 and 0.25 (Pardo et al., 2010) to facilitate boundary application on management practices. Overall, the final classification from the INVMB indicated that the percentage of samples in the different status classes was 23.7% reference sites, 8.1% (“High”), 12.8% (“Good”), 22.4% (“Moderate”), 20.8% (“Poor”) and 12.2% (“Bad”). All sites denoted as references showed EQR values higher than 0.7, which indicated that the distribution of these sites was always within the “High” and “Good” quality classes.

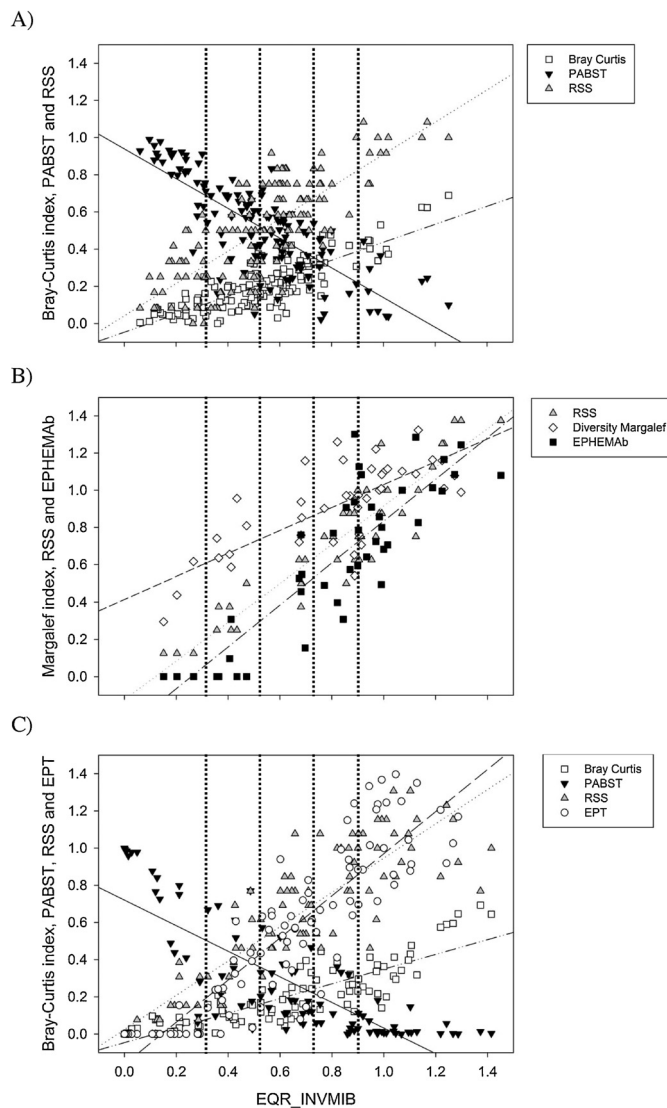


Fig. 5. Bivariate graphic between the selected metrics and the EQR_INVMB in each stream type: (a) lowland streams, (b) canyons and (c) mountain streams. Final boundaries between different ecological status classes (0.93, 0.73, 0.50 and 0.25) are also included in the figure as vertical discontinuous lines.

3.5. Ecological responses across status classes

The five resulting ecological status classes derived from our multimetric index assessment represented the whole range of the existing variation in hydromorphological parameters and land uses across stream types. The final values for the individual metrics, the INVMB and EQR_INVMB per ecological quality class, and for the reference sites, in each stream type were summarised in Table 5. These values evidenced a clear degradation from the best classes (“High” and “Good”) to the worst ecological-status classes (“Poor” and “Bad”). At the same time, the “Moderate” class overlapped in some cases with other status classes. All multimetric indices showed lower variability than the individual metrics (Table 5).

Land uses and water chemistry confirmed the degradation evidenced by the individual metrics and the multimetric indices (Table 6). In lowland streams, the nutrient enrichment was clearly illustrated by an increase of nutrients (NH_4 and PO_4) and periphytic biomass (Chl a), which indicated impairment from sewage effluents and agriculture activity. Oxygen was strongly reduced only in the worst quality class (Table 6). In canyon streams, water chemical composition showed lower levels of impact, independently of

Table 5
Values (mean $\pm$ SD) of the individual metrics used to obtain the multimetric index (INVMIB) and its EQR in each of the five ecological quality classes per stream type. Coefficients of variation are shown between brackets. Values of these metrics are also presented for our reference sites.

Multimetric index	References	High ^a	Good	Moderate	Poor	Bad
Lowlands (INV1MIB)	Bray Curtis PABST RSS INV1MIB EQR INV1MIB	44.64 $\pm$ 0.16 (0.4) 0.24 $\pm$ 0.06 (25.0) 10.0 $\pm$ 1.0 (10.0) 2.83 $\pm$ 0.01 (3.6) 0.94 $\pm$ 0.01 (1.1)	33.99 $\pm$ 3.22 (9.5) 0.3 $\pm$ 0.05 (16.7) 8.01 $\pm$ 0.56 (7.0) 2.33 $\pm$ 0.03 (1.3) 0.78 $\pm$ 0.01 (1.3)	23.4 $\pm$ 1.24 (5.3) 0.43 $\pm$ 0.02 (4.7) 6.8 $\pm$ 0.33 (4.9) 1.81 $\pm$ 0.04 (3.4) 0.6 $\pm$ 0.01 (1.7)	14.42 $\pm$ 1.16 (8.0) 0.62 $\pm$ 0.02 (3.2) 4.29 $\pm$ 0.43 (10.0) 1.18 $\pm$ 0.04 (3.4) 0.39 $\pm$ 0.01 (2.6)	5.88 $\pm$ 0.88 (15.0) 0.89 $\pm$ 0.01 (1.1) 2.95 $\pm$ 0.37 (12.5) 0.52 $\pm$ 0.04 (7.7) 0.17 $\pm$ 0.01 (5.9)
Canyons (INV2MIB)	EPHEMAB Margalef index RSS INV2MIB EQR INV2MIB	2579.8 $\pm$ 1804.77 (70.0) 4.46 $\pm$ 0.38 (8.5) 6.2 $\pm$ 0.49 (7.9) 2.82 $\pm$ 0.07 (2.5) 1.01 $\pm$ 0.03 (3.0)	240.33 $\pm$ 203.21 (84.6) 4.53 $\pm$ 0.41 (9.1) 5.33 $\pm$ 0.88 (16.5) 2.38 $\pm$ 0.05 (2.1) 0.86 $\pm$ 0.02 (2.3)	84 $\pm$ 70.81 (84.3) 4.00 $\pm$ 0.49 (12.1) 4.00 $\pm$ 0.58 (14.5) 1.91 $\pm$ 0.01 (0.5) 0.69 $\pm$ 0.01 (1.4)	1.29 $\pm$ 1.13 (87.6) 3.03 $\pm$ 0.21 (6.9) 2.43 $\pm$ 0.37 (15.2) 1.08 $\pm$ 0.07 (6.5) 0.39 $\pm$ 0.02 (5.1)	– 1.56 $\pm$ 0.30 (19.2) 1.00 $\pm$ 0.00 (0.0) 0.49 $\pm$ 0.07 (14.3) 0.18 $\pm$ 0.03 (16.7)
Mountains (INV5MIB)	Bray Curtis PABST RSS EPT index INV5MIB EQR INV5MIB	43.45 $\pm$ 8.97 (20.6) 0.03 $\pm$ 0.01 (33.3) 8.75 $\pm$ 1.11 (12.7) 30.06 $\pm$ 2.64 (8.6) 4.32 $\pm$ 0.39 (9.0) 1.09 $\pm$ 0.10 (9.2)	23.78 $\pm$ 2.7 (11.4) 0.17 $\pm$ 0.04 (23.5) 9.6 $\pm$ 0.81 (8.4) 15.82 $\pm$ 0.61 (3.9) 3.04 $\pm$ 0.08 (2.6) 0.76 $\pm$ 0.02 (2.6)	14.68 $\pm$ 2.13 (14.5) 0.29 $\pm$ 0.03 (10.3) 8.59 $\pm$ 0.58 (6.8) 11.62 $\pm$ 1.18 (10.2) 2.43 $\pm$ 0.05 (2.1) 0.61 $\pm$ 0.01 (1.6)	9.40 $\pm$ 1.58 (16.8) 0.35 $\pm$ 0.06 (17.1) 4.27 $\pm$ 0.67 (15.7) 4.95 $\pm$ 1.09 (22.0) 1.55 $\pm$ 0.07 (4.5) 0.39 $\pm$ 0.02 (5.1)	3.74 $\pm$ 0.88 (23.5) 0.81 $\pm$ 0.05 (6.2) 1.13 $\pm$ 0.39 (34.5) 0.23 $\pm$ 0.03 (7.9) 0.38 $\pm$ 0.08 (21.1) 0.10 $\pm$ 0.02 (20.0)

^a High status class does not include our reference sites.

the status class, although we can observe an increase of nitrates from those sites included in the higher quality classes to those sites included in the worst status (Table 6), as well as some differences regarding the percentage of land uses within this stream type. Finally, impacts in mountain streams were reflected by a strong increase of nutrient concentrations (i.e., NO₃ and PO₄) and artificial land uses, together with a reduction of oxygen values from “Moderate” to “Bad” classes, indicating high levels of eutrophication (Table 6). These results together with the ordinations of sites per stream type (Appendix C) reflected a general degradation of the studied sites from the highest quality status classes to the most impaired sites and with a gradual differentiation among the ecological quality classes.

4. Discussion

The INVMIB based on invertebrates is a new multimetric classification system fulfilling the scientific requirements of the European legislation (WFD; European Commission, 2000). The multimetric index has been developed using the most extensive monitoring network of invertebrate communities in the Balearic Islands, 60 studied streams that included a total of 29 river basins (i.e., 73.7% of the Islands temporary streams). It is known that invertebrate succession, when water resumes after the dry summer, advances from the existing pool of species inhabiting subterranean springs, hyporheos, resistant forms to drought and from aerial colonisation from nearby water bodies to temporary streams (Williams and Hynes, 1976; Alvarez and Pardo, 2007). These resistant taxa, usually of small size, develop in very short time and increase in abundance towards the end of the water period (Pardo and Álvarez, 2006; Alvarez and Pardo, 2005, 2009), unless sudden high discharge levels, caused by rains, disturb the communities and reduce the numbers (Lake, 2003; Boulton and Lake, 2008). We selected a sampling period from November to May during the years of study (avoiding late autumnal extended droughts and late spring floods), that ensured the establishment of the typical invertebrate community in each stream type, providing a strong basis for a reliable biological assessment system. Indeed, the benefit of using datasets from different seasons to achieve a better categorisation and prediction of invertebrate communities has been previously demonstrated for temperate streams (Furse et al., 1984; Sporka et al., 2006), even though the sampling period proposed in this study for temporary streams is highly restricted to winter–spring months.

The sampling design, timing and methodology, here applied, allowed us to successfully evaluate the natural variability of the invertebrate communities characterising the temporary streams. Our study revealed that the invertebrate communities under reference conditions varied with the geomorphological setting and water origin. The analyses of the invertebrate reference assemblages confirmed our *a priori* classification into the 3 types: lowland, canyon and mountain streams. Reference canyon streams were the most heterogeneous group, which is attributed to the characteristics of the life cycles of the resident species adapted to this particular habitat since they are disconnected from continuous spring sources and remain temporally isolated as water ponds or “gorgs” (Williams and Hynes, 1976). Indeed, those canyon streams which usually dry up showed similar communities among them, while others with continuous flowing water were similar to the mountain streams. Invertebrate assemblages characterising mountain and lowland streams were distinctive and showed a high dissimilarity among stream types. Therefore, invertebrate communities are stream-type specific, similar to that observed in the transitional coastal lagoons in the Balearic Islands where invertebrate communities allowed a clear differentiation among types (Lucena-Moya et al., 2009).

Table 6Physico-chemical characteristics, hydromorphological parameters and land uses of sampled sites included in the reference sites and in each of the five ecological quality classes per stream type (mean $\pm$ SD).

Stream type	Status classes	Physicochemical variables							
		EC (μ S/cm)	pH	P.PO ₄ (mg/l)	N.NO ₃ (mg/l)	N.NH ₄ (mg/l)	Oxygen (mg/l)	Chlorophyll <i>a</i> (mg/m ²)	AFDM (g/m ²)
Lowlands	Reference	747.86 $\pm$ 29.46	7.66 $\pm$ 0.12	0.01 $\pm$ 0.01	0.32 $\pm$ 0.13	0.03 $\pm$ 0.02	9.36 $\pm$ 0.56	5.92 $\pm$ 2.23	6.18 $\pm$ 1.07
	High	1574.0 $\pm$ 566.00	7.97 $\pm$ 0.30	0.00 $\pm$ 0.00	2.04 $\pm$ 0.29	0.03 $\pm$ 0.03	7.95 $\pm$ 2.70	7.10 $\pm$ 5.63	6.28 $\pm$ 4.16
	Good	1022.68 $\pm$ 149.52	7.94 $\pm$ 0.11	0.07 $\pm$ 0.03	1.38 $\pm$ 0.42	0.01 $\pm$ 0.0	9.77 $\pm$ 0.52	17.00 $\pm$ 4.75	9.70 $\pm$ 2.17
	Moderate	1544.56 $\pm$ 95.44	7.91 $\pm$ 0.05	0.29 $\pm$ 0.12	2.79 $\pm$ 0.71	0.14 $\pm$ 0.06	9.06 $\pm$ 0.37	20.26 $\pm$ 2.95	13.42 $\pm$ 1.41
	Poor	1862.42 $\pm$ 108.02	7.82 $\pm$ 0.07	0.37 $\pm$ 0.12	4.28 $\pm$ 0.84	0.30 $\pm$ 0.22	8.09 $\pm$ 0.46	20.43 $\pm$ 3.53	13.81 $\pm$ 1.58
	Bad	1437.11 $\pm$ 170.69	7.75 $\pm$ 0.10	1.19 $\pm$ 0.36	2.00 $\pm$ 0.64	1.50 $\pm$ 0.51	6.14 $\pm$ 0.90	24.88 $\pm$ 5.39	27.64 $\pm$ 6.20
Canyons	Reference	387.44 $\pm$ 41.33	8.17 $\pm$ 0.09	0.02 $\pm$ 0.01	0.34 $\pm$ 0.13	0.02 $\pm$ 0.01	10.15 $\pm$ 0.51	9.01 $\pm$ 2.44	9.88 $\pm$ 2.14
	High	416.23 $\pm$ 86.52	8.02 $\pm$ 0.21	0.02 $\pm$ 0.01	0.22 $\pm$ 0.07	0.03 $\pm$ 0.02	8.11 $\pm$ 0.82	13.01 $\pm$ 2.75	20.56 $\pm$ 7.34
	Good	397.68 $\pm$ 62.03	8.12 $\pm$ 0.17	0.02 $\pm$ 0.02	0.18 $\pm$ 0.13	0.03 $\pm$ 0.01	9.54 $\pm$ 0.55	7.14 $\pm$ 2.25	4.69 $\pm$ 2.04
	Moderate	578.50 $\pm$ 53.35	7.78 $\pm$ 0.32	0.00 $\pm$ 0.00	0.3 $\pm$ 0.15	0.00 $\pm$ 0.00	9.25 $\pm$ 0.41	7.43 $\pm$ 3.41	13.41 $\pm$ 9.85
	Poor	615.09 $\pm$ 29.89	7.85 $\pm$ 0.22	0.03 $\pm$ 0.01	0.9 $\pm$ 0.21	0.02 $\pm$ 0.01	8.91 $\pm$ 0.32	15.46 $\pm$ 3.27	24.07 $\pm$ 6.57
	Bad	660.00 $\pm$ 122.50	8.11 $\pm$ 0.37	0.02 $\pm$ 0.02	1.41 $\pm$ 0.22	0.00 $\pm$ 0.00	8.45 $\pm$ 0.76	6.54 $\pm$ 6.15	19.33 $\pm$ 16.20
Mountains	Reference	753.68 $\pm$ 55.5	8.04 $\pm$ 0.06	0.02 $\pm$ 0.0	0.54 $\pm$ 0.12	0.02 $\pm$ 0.01	9.65 $\pm$ 0.19	7.24 $\pm$ 1.25	7.73 $\pm$ 1.83
	High	552.50 $\pm$ 60.30	8.16 $\pm$ 0.18	0.02 $\pm$ 0.01	0.65 $\pm$ 0.36	0.00 $\pm$ 0.00	9.99 $\pm$ 0.20	18.4 $\pm$ 8.24	5.46 $\pm$ 1.02
	Good	603.05 $\pm$ 78.96	7.83 $\pm$ 0.14	0.02 $\pm$ 0.01	0.70 $\pm$ 0.27	0.01 $\pm$ 0.00	9.93 $\pm$ 0.93	27.07 $\pm$ 8.30	14.15 $\pm$ 4.53
	Moderate	704.36 $\pm$ 57.46	7.79 $\pm$ 0.11	0.03 $\pm$ 0.01	0.81 $\pm$ 0.14	0.06 $\pm$ 0.03	9.59 $\pm$ 0.55	21.79 $\pm$ 6.56	13.81 $\pm$ 2.79
	Poor	940.16 $\pm$ 85.18	7.62 $\pm$ 0.10	0.13 $\pm$ 0.07	1.28 $\pm$ 0.36	0.03 $\pm$ 0.01	7.81 $\pm$ 0.65	17.5 $\pm$ 3.27	12.73 $\pm$ 3.11
	Bad	1121.34 $\pm$ 83.57	7.85 $\pm$ 0.07	0.95 $\pm$ 0.33	1.58 $\pm$ 0.46	0.72 $\pm$ 0.39	5.30 $\pm$ 0.84	16.25 $\pm$ 4.36	17.0 $\pm$ 2.97
Stream type	Status classes	Hydromorphological parameters and land uses							
		HQA score	HMS score	Artificial (%)	Agriculture (%)	Forestry (%)			
Lowlands	Reference	36.0 $\pm$ 2.67	2.0 $\pm$ 0.67	0.0 $\pm$ 0.0	18.01 $\pm$ 3.78	81.99 $\pm$ 3.78			
	High	34.50 $\pm$ 0.50	44.00 $\pm$ 41.00	4.71 $\pm$ 4.17	52.28 $\pm$ 3.87	43.02 $\pm$ 8.04			
	Good	39.27 $\pm$ 3.18	30.23 $\pm$ 5.88	1.32 $\pm$ 0.40	31.25 $\pm$ 6.75	58.46 $\pm$ 5.36			
	Moderate	37.51 $\pm$ 1.37	35.03 $\pm$ 4.05	1.60 $\pm$ 0.27	49.26 $\pm$ 2.60	46.65 $\pm$ 2.08			
	Poor	31.04 $\pm$ 1.62	28.64 $\pm$ 5.24	1.41 $\pm$ 0.35	53.29 $\pm$ 4.63	41.35 $\pm$ 3.70			
	Bad	31.73 $\pm$ 2.00	45.73 $\pm$ 6.60	1.86 $\pm$ 0.19	57.39 $\pm$ 5.20	40.75 $\pm$ 5.27			
Canyons	Reference	38.94 $\pm$ 0.41	12.0 $\pm$ 3.9	0.66 $\pm$ 0.25	6.92 $\pm$ 2.49	92.42 $\pm$ 2.74			
	High	37.50 $\pm$ 5.36	33.20 $\pm$ 3.32	1.23 $\pm$ 0.80	23.05 $\pm$ 2.49	75.72 $\pm$ 3.24			
	Good	43.33 $\pm$ 3.83	33.33 $\pm$ 5.33	0.75 $\pm$ 0.75	20.80 $\pm$ 0.77	78.46 $\pm$ 1.51			
	Moderate	25.17 $\pm$ 7.17	34.67 $\pm$ 3.33	2.61 $\pm$ 1.31	28.57 $\pm$ 4.27	68.81 $\pm$ 5.58			
	Poor	33.43 $\pm$ 5.45	16.29 $\pm$ 7.68	1.68 $\pm$ 0.79	29.97 $\pm$ 1.02	68.35 $\pm$ 1.81			
	Bad	39.50 $\pm$ 5.50	39.50 $\pm$ 9.50	1.96 $\pm$ 1.96	30.33 $\pm$ 2.52	67.71 $\pm$ 4.48			
Mountains	Reference	45.8 $\pm$ 1.8	15.33 $\pm$ 3.7	0.0 $\pm$ 0.0	6.33 $\pm$ 0.45	93.67 $\pm$ 0.45			
	High	38.50 $\pm$ 0.00	77.00 $\pm$ 0.00	1.53 $\pm$ 0.00	17.54 $\pm$ 0.00	80.93 $\pm$ 0.00			
	Good	34.35 $\pm$ 4.34	25.0 $\pm$ 1.58	1.18 $\pm$ 0.26	20.37 $\pm$ 4.66	73.19 $\pm$ 2.30			
	Moderate	32.59 $\pm$ 2.37	35.85 $\pm$ 6.69	1.79 $\pm$ 0.48	22.7 $\pm$ 3.37	69.80 $\pm$ 1.69			
	Poor	40.67 $\pm$ 3.14	20.63 $\pm$ 5.26	3.08 $\pm$ 0.64	18.00 $\pm$ 3.71	69.47 $\pm$ 1.85			
	Bad	40.94 $\pm$ 1.74	38.78 $\pm$ 4.85	3.90 $\pm$ 0.58	27.62 $\pm$ 1.32	68.48 $\pm$ 1.61			

The INVMIB consisted of the combination of individual metrics for each stream type (Heiskanen et al., 2004) covering structural parameters from WFD Annex V (abundance, diversity, species composition, and tolerance/sensitivity), offering a more reliable tool than assessment methods based on single metrics (Barbour et al., 1995, 1999; Birk et al., 2012). The potential of each individual metric was evaluated according to its relevance and its ability to discriminate among sites (Barbour et al., 1999; Hering et al., 2003, 2006b; Ofenböck et al., 2004; Bennett et al., 2011). Diversity/richness measures contributed to the three multimetric indices developed in this study, as well as tolerance/sensitivity metrics that showed a high discriminatory power. The invertebrate communities inhabiting the studied temporary streams were characterised by many sensitive and tolerant taxa (Appendix B). Some invertebrates were identified as sensitive taxa in more than one stream type (e.g., *Agabus*, *Hydroptila*, *Cloeon*, *Laccobius*, *Meladema*, *Caenis*, *Simulium*, *Hydroporus*, *Stictometes* and *Tinodes*), while others were identified as tolerant taxa in more than one stream type (e.g., *Proasellus*, *Chironomus*, *Physella*, *Tipula*, nematodes and oligochaetes).

Despite there are many multimetric approaches based on quantitative data in the Mediterranean area (e.g., Buffagni et al., 2005, 2006; Erba et al., 2009; Inag, 2009; Munné and Prat, 2009), the assessment of temporary streams has not been specifically addressed as the particular water bodies they are, with comprehensive designed databases such as the one in this study. The invertebrate multimetric indices estimated in this study strongly responded to multiple degradation pressures influencing streams of the Balearic Islands while providing an integrated view of the biological community of a site (Karr and Chu, 1999; Hering et al., 2006b). In general, the multimetric indices correlated better with the general degradation gradient, which includes many different impacts, than with the type-specific organic contamination gradient, except for mountain streams which showed strong responses to both general and organic degradation pressures. This fact evidenced that invertebrates are very sensitive to many different impacts such as the lack of riparian vegetation, hydromorphological alterations and/or substrate type. Despite the relationships between the EQR and the general degradation gradient are significant, they were not too strong, especially because the construction of a degradation gradient by using PCA introduces variation (i.e., combination of environmental variables), and not every taxon responds to the same environmental variables (Verdonschot et al., 2012). Nonetheless, these values are comparable to the response of other multimetric indices or individual macroinvertebrate metrics to a PCA-gradient ($R^2=0.09\text{--}0.37$; see Verdonschot et al., 2012; Johnson et al., 2006).

Our results allowed to validate the classification boundaries established a priori since the interpretation of crossing points was indicative – following proposed approaches to identify boundaries in linear responses to pressures (Pollard and van de Bund, 2005) – of major changes in community composition, difficult to recover. Individual metrics and multimetric indices showed low variability within reference biological conditions, strong significant responses to pressures and discriminatory efficiencies that allowed a good differentiation among reference and non-reference sites. For the canyon streams, the lower number of reference sites and the fact that none of the sites was strongly degraded by pollution, together with its particular hydrological characteristics, limited the power of the analyses, needing further future data to test the results obtained in this study.

The classification system obtained with the INVMIB reflected the ecological quality of temporary streams of Balearic Islands in accordance with corresponding values of physico-chemical variables, hydromorphological parameters and land uses. A wide spectrum of ecological status was identified from the “High” class (sites with high EQR values), to “Poor” and “Bad” classes (sites with low EQR

values). Lowland streams were more affected, at both biological and physicochemical levels (i.e., higher values of nitrates, ammonium, phosphate and chlorophyll *a*), than mountain streams and canyons, due to their location within the most populated areas. On the other side, canyons showed a particular fauna with special behavioural or physiological adaptations that tend to survive during the dry season by colonising “gorgs” (Williams and Hynes, 1977; Marchant, 1982). Canyons are located in remote areas so they suffered few impacts, with the exception of some sites affected by punctual impacts from oil spills or organic effluents. In addition, canyoning is a common practise that involves climbing, scrambling and rappelling, something that we could not evaluate in this study as people has open access to them. Mountain streams covered a large gradient of degradation from natural areas (i.e., low nutrient values, modest agriculture practises and without important channel alterations) to impacted sites receiving mainly local inputs from urban wastewater and suffering increasing water abstraction for urbanisation and agriculture uses. Some mountain streams classified in the “Good” ecological status class showed high HMS scores, which could be a negative influence. However, these high values are due to the existence of bank reinforcement structures made with typical stone walls (“martges”) of porous material only to avoid erosion but that ensure the lateral connectivity between the stream and its floodplain, a critical connexion to both ecosystem structure and functioning (Ward, 1989).

5. Conclusion

Over the last decade, in accordance with the WFD, various multimetric indices have been developed, tested and suggested for the classification of the ecological status. Despite a good application on permanent streams, attempts to evaluate the ecological quality of temporary streams are scarce and not type-specific. The spatial and temporal variability of these ecosystems impose challenges to the assessment of communities. The utilisation of a sound network sampling design, including different stream types and the natural variability of reference conditions, allowed us to develop reliable tools sensitive to human impacts for the assessment of temporary streams. EQRs from winter and spring were strongly correlated with the mean EQR, and thus we conclude one of these sampling seasons as the best season to sample temporary streams, avoiding multi-seasonal sampling to reduce efforts and costs. The new multimetric system fulfilled all requirements of a classification system, in a similar way as other classification systems developed for permanent streams. This new assessment system is currently being applied by the water authorities of the Balearic Islands.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ecolind.2014.05.030>.

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